

Order Scorpiones

Scorpions

The most ancient group of all arachnids, scorpions are recognizable by their large, clawlike pedipalps, and their sting-bearing tails. They hunt after dark, often finding prey by touch. Scorpions sometimes use their stings to paralyze prey, but the sting's key function is defense. Females give birth to live young, which they carry on their backs. There are about 1,500 species in 15 families.

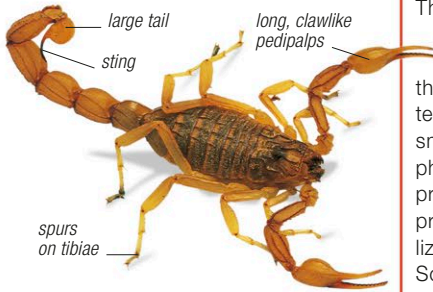
Family Buthidae

Buthids

Occurrence c.650 spp. worldwide, in warmer regions; in rock cracks, under stones, logs, bark



Despite their relatively small size, 3¼–4¾ in (8–12 cm) in length, buthids are the most dangerous scorpions. Their potent venom can paralyze the heart and the respiratory system.



BUTHUS OCCITANUS
An inhabitant of scrubby ground, the common yellow scorpion occurs in parts of North Africa and West Asia. Antidotes to its venom are available.

Family Scorpionidae

Scorpionids

Occurrence 196 spp. in Africa, Asia, and Australia; in cracks, caves, under stones, logs

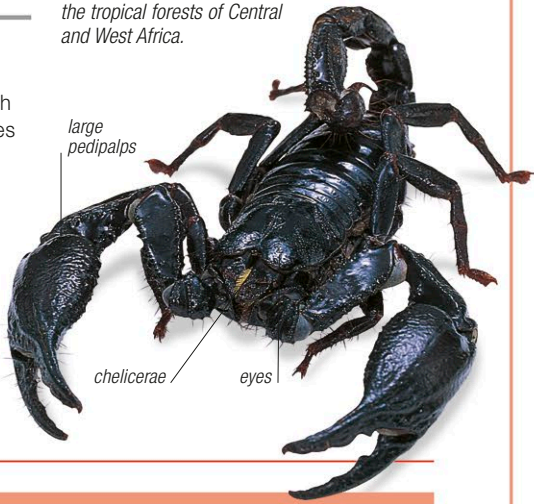


This family includes the world's largest scorpions, measuring up to 8½ in (21 cm) long. Although they look menacing, the larger species tend to be less venomous than the smaller ones and rely primarily on physical strength to overpower their prey. Many of these dig to locate prey, which could include spiders, lizards, and even small mammals. Scorpionids sting not only to paralyze their victim, but may also sting mates as part of their complex mating ritual. A litter of 30–35 young stay with the mother

for some time after their first molt. Scorpionids have a 5-sided sternum in contrast to the buthids' triangular one.

PARDINUS IMPERATOR

This is the emperor scorpion, some specimens of which can exceed 2½ oz (60 g) in weight. It can be found among leaf litter in the tropical forests of Central and West Africa.



Order Pseudoscorpiones

Pseudoscorpions

Also called false scorpions because they lack a sting and tail, members of this order of 25 families (about 3,300 species) are similar to scorpions in shape. They have venom glands in pincerlike parts in their pedipalps and prepare silk nests in which to molt, brood their young, and hibernate. Eggs are laid in a special brood pouch. Females may cling to other animals to disperse to new habitats.

Family Neobisiidae

Neobisiids

Occurrence c.500 spp. worldwide, especially in the Northern Hemisphere; in leaf litter, soil, caves

Habitat Terrestrial

The members of this family are very small, measuring ½–¾ in (1–5 mm) in length. They usually have 4 eyes but cave-living species may have fewer eyes or none at all. In color, neobisiids vary from olive- to dark brown, with red, yellow, or cream tinges. The legs are often slightly green, with 2-segmented tarsi. The claw of each pedipalp has one venom gland located in the fixed finger; after paralyzing its prey, the neobisiid shreds it with its large mouthparts.



NEOBISIUM MARITIMUM
Measuring ⅛–¼ in (3–3.5 mm) long, the maritime pseudoscorpion is found in cracks in rocks and under stones in coastal areas of Ireland, England, and France.

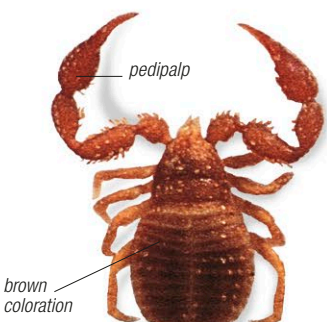
Family Cheliferidae

Cheliferids

Occurrence c.300 spp. worldwide, especially in warmer regions; in leaf litter, on tree bark



Unlike in neobisiids, in cheliferids both fingers of the pincers have glands that produce the venom they use to paralyze their prey. There are no teeth on the inner surfaces of the pincers. Cheliferids are ¾–¾ in (1.5–5 mm) long and usually have 2 eyes. Coloration varies from pale to dark brown or black, often tinged with red or olive, with dark markings. As in many arachnids, courtship can be complex; males and females dance holding each other's pedipalps.



DACTYLOCHELIFER SP.
While most of these species are distributed across the Northern Hemisphere in a variety of habitats, some are confined to coastal habitats.

Order Solifugae

Sun-spiders or wind-scorpions

Neither true spiders nor scorpions, the animals in this order lack poison glands. Instead, they kill and chew up their prey with their very large, pincerlike chelicerae. Their leglike pedipalps have suction pads that hold down the prey so that it can be eaten. There are about 1,100 species, in 12 families, and most are found either in North America or in the tropics.

Family Solpugidae

Solpugids

Occurrence 190 spp. in Africa, parts of West Asia; on ground, under stones, logs, in burrows in sandy soil



Pale, brown or yellowish, sometimes with bright markings, these sun-spiders are ¼–2¼ in (0.6–6 cm) in length. Their 4 pairs of legs have a variable number of tarsi and all, except the first pair, have claws with a smooth surface. They have forward-facing chelicerae to kill prey; small species and nymphs eat termites. Females lay masses of

20–100 eggs. Some solpugids are active by day; others burrow in sandy soil or hide in cracks, or under stones during the day, and forage in tree canopies at night.



METASOLPUGA PICTA
This species is found in the Namib Desert in Namibia. Like all solpugids, it will bite if handled carelessly, but does not have venom glands.

Family Ammotrechidae

Ammotrechids

Occurrence 83 spp. in the warmer regions of South, Central, and North America; in rotten tree stumps



This family comprises relatively slender species that measure ½–¾ in (0.4–2 cm) in length. They are colored in shades of brown. The front margin of the head is rounded, and the tarsi of the first pair of legs do not have claws. Many species are nocturnal and dig into the soil during the day. Some of the smaller ones hide in termite galleries

or inside the tunnels of wood-boring insects. These arachnids are predatory on a range of arthropods. Females may lay 2 or more masses of 20–150 eggs in their lifetime.



AMMOTRECHELLA STIMPSONI
This species is found in Florida and other areas in southern North America, where it lives in the bark of rotten, termite-infested trees.